Auteur: Thomas Olszowka
Studierichting: Informatica
Oefeningenreeks 3A nr. 36.16

$$
\begin{aligned}
f(x) &= \ln(x^3 + \sqrt{x^6+1}) \\
f(x) &= \ln(g(x)) \text{ met } g(x) = x^3 + \sqrt{x^6+1} \\
f'(x) &= \frac{g(x)'}{g(x)} \\
&= \frac{g(x)'}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 + (\sqrt{x^6+1})'}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 + \frac{1}{2\sqrt{x^6+1}}6x^5}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 + \frac{6x^5}{2\sqrt{x^6+1}}}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 + \frac{3x^5}{\sqrt{x^6+1}}}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 \left(1 + \frac{x^3}{\sqrt{x^6+1}}\right)}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 \left(\frac{\sqrt{x^6+1}}{\sqrt{x^6+1}} + \frac{x^3}{\sqrt{x^6+1}}\right)}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2 \left(\frac{x^3 + \sqrt{x^6+1}}{\sqrt{x^6+1}}\right)}{x^3 + \sqrt{x^6+1}} \\
&= 3x^2 \left(\frac{x^3 + \sqrt{x^6+1}}{\sqrt{x^6+1}}\right) \frac{1}{x^3 + \sqrt{x^6+1}} \\
&= \frac{3x^2}{\sqrt{x^6+1}}
\end{aligned}
$$

References